

Fully nonlinear neuromorphic computing with linear wave scattering

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I. THE SCATTERING MATRIX CAN APPROXIMATE ARBITRARY NON-LINEAR FUNCTIONS

Here, we prove for the case of scalar input and output that the scattering matrix of a system of coupled modes can be used to approximate arbitrary analytic functions $x \rightarrow f(x) \in \mathbb{R}$. The scalar input $x \in \mathbb{R}$ is encoded in the detunings of each mode and the function is approximated by optimizing (some of) the other system parameters. In the limit of an infinitely large system, the approximation becomes exact.

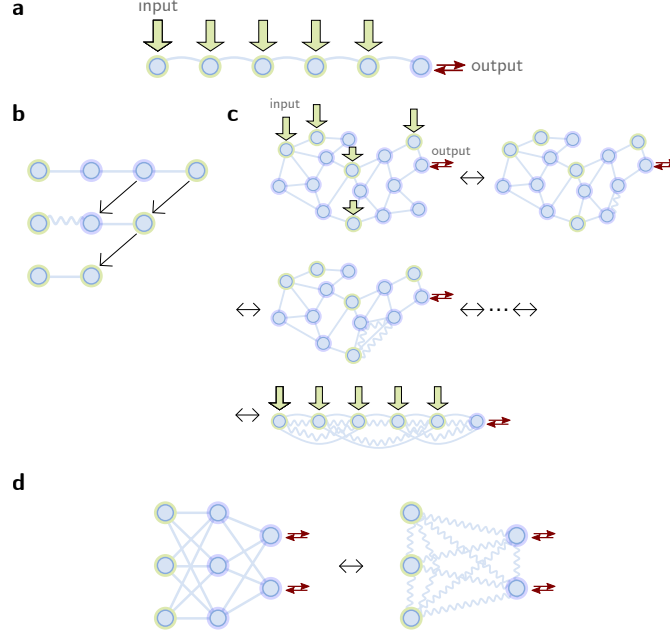


FIG. S1. **Mapping of a multimode system to a one-dimensional chain.** **a** The scattering matrix of a one-dimensional chain with nearest-neighbour couplings in which the input enters in the detuning of each mode has the form of a generalised continued fraction which can be used to approximate any analytic function. In the limit of an infinite chain length, this approximation approaches the exact result. **b** Reduction rules to map between systems of coupled modes [1]. Eliminating an intermediate mode in a chain of modes coupled via beamsplitter couplings results in effective dissipative coupling with added local decay (wiggly lines), Eq. (S3), a further elimination step yields effective coherent beamsplitter coupling with added decay (straight lines), Eq. (S4). This technique is discussed in more detail and generalised to combinations of beamsplitter and two-mode-squeezing couplings in Ref. [1]. **c** Employing the reduction rules of **b**, one can map *any* system of arbitrarily coupled modes to a one-dimensional chain with couplings of varying range. For clarity, not all resulting effective couplings are shown. **d** Any neuromorphic system with input dimension D and input replication R , e.g., the three-layer system of Fig. 2 a, can be mapped to a system consisting only of the input and output modes with all-to-all dissipative couplings. The number of these dissipative couplings equals the number of couplings we can at most control independently in a neuromorphic system with this input dimension and input replication.

A. One-dimensional chain with nearest-neighbour couplings

The Green's function of a one-dimensional chain with nearest-neighbour couplings is particularly straightforward to compute. We consider the setup for a chain of length $N = R + 1$, with R the number of input replications, shown in Fig. S1 a. The detuning at the j th mode is given by $\Delta_j + x$ for $j \leq N = R + 1$ and nearest-neighbour modes j and $j + 1$ are coupled at strength J_j . The system is probed at the N th site where the detuning is simply Δ_N . Similar to the derivation of the scattering matrix (15) for the layered architecture, we consider the equation of motions in the frequency domain

$$\begin{aligned}
 \omega a_j(\omega) &= \left(-i \frac{\kappa_j}{2} + (\Delta_j + x) \right) a_j + J_{j-1} a_{j-1} + J_{j+1} a_{j+1}, \\
 \omega a_N(\omega) &= \left(-i \frac{\kappa_N + \kappa'_N}{2} + (\Delta_N + x) \right) a_N + J_{N-1} a_{N-1} \\
 &\quad - \sqrt{\kappa'_N} a_{N,\text{probe}}
 \end{aligned} \tag{S1}$$

discussed in more detail and generality in the Supplementary Information of Ref. [1] where also the generality of the argument to two-mode and single-mode squeezing is derived; a related technique is employed in Ref. [3, 4].

Next, we show that a system or arbitrary architecture can be used to approximate arbitrary analytic, scalar functions arbitrary well. For the following arguments, the values of the effective coupling strength, effective local decay and detuning shifts are not relevant. We merely employ this reduction technique to map one physical system to one that is easier to analyse. In particular, we map the generic system of multiple modes, such as the first panel in Fig. S1 c, to a chain with arbitrary, possibly long-range couplings, such as the chain in the last panel of Fig. S1 c, consisting only of the modes whose detunings encode the input and the mode p which is used to probe the system which is the $(R+1)$ th mode of the chain. This is possible by performing multiple reduction steps eliminating all other modes as is sketched in the other panels of Fig. S1 c. The couplings between the modes of this reduced model are now a combination of coherent beamsplitter coupling and dissipative coupling, with coupling constants which may even be complex. We introduce the effective coupling strengths $g_{j,\ell}$ including all of these terms for the effective coupling between modes j and ℓ . We note that $g_{j,\ell}$ need not be symmetric, i.e., $g_{j,\ell} \neq g_{\ell,j}^*$. The on-site terms, i.e., decay rate and detuning, were also modified by the reduction to the chain and we define $\mu_j \equiv -\frac{\kappa_j}{2} - i\Delta_j$ as the effective on-site term of the j th site in the equations of motion for the reduced chain. We can then proceed as before for the chain with nearest-neighbour couplings to derive the scattering matrix element at the $(R+1)$ th site, $S_{R+1,R+1} = S_{p,p}$ by eliminating mode by mode along the chain

$$S_{R+1,R+1}(\omega, x, \theta) = 1 + \frac{\kappa_{R+1}}{\tilde{\mu}_{R+1}} \quad (\text{S5})$$

with

$$\tilde{\mu}_j = \mu_j + \frac{g_{j,j-1}g_{j-1,j}}{\tilde{\mu}_{j-1}} + \sum_{\substack{\text{paths of} \\ \text{length } \ell: \\ 3 \leq \ell \leq 2^j}} (-i)^\ell \frac{g_{j,m_1}g_{m_1,m_2} \cdots g_{m_\ell,j}}{\tilde{\mu}_{m_1}\tilde{\mu}_{m_2} \cdots \tilde{\mu}_{m_\ell}} \quad (\text{S6})$$

with $\tilde{\mu}_1 = \mu_1$. Here, the sum goes over all paths from mode j to modes $m_1, m_2, \dots, m_\ell < j$ and back to mode j that have a length smaller than 2^j and involve at least one coupling of a range longer than one, i.e., next-nearest neighbour coupling and beyond. Note that the scalar input x enters as summand in each μ_j .

As a result of this exercise, we see that Eq. (S5) reduces to Eq. (S2) in the case of only symmetric nearest-neighbour couplings in which case we know that $S_{R+1,R+1}(\omega, x, \theta)$ can represent arbitrary, non-linear analytic functions. In the general case, the expression for $S_{R+1,R+1}(\omega, x, \theta)$, Eq. (S5), contains sums of generalised continued fractions in which the coefficients are again generalised continued fractions. So Eq. (S5) has both more parameters and contains more general terms, e.g., powers of x , so is by no means more restrictive than expression (S2) which already allowed to approximate arbitrary analytic functions. We therefore conclude that a scattering matrix of any linear system in which the input enters R times can be used to approximate arbitrary analytic, scalar functions. The number of input replications R determines the recursive depth of expression (S5) and therefore plays a crucial role for the accuracy of the approximation.

To exemplify the expressive power of the scattering matrix, we employ the system sketched in Fig. S2 a to fit different non-linear functions ranging from a combination of hyperbolic tangents to oscillating functions. The system we consider consists of a first layer with R modes in which the input is replicated R times in the detunings $\Delta_j^{(1)} + x$ of the first layer, then a second layer of variable size and an output layer with only one mode. The decay rate κ is considered equal at all sites.

We note that functions with sharper features or many oscillations require larger R than smoother functions. Furthermore, we selected for each plot the largest value of κ that still yielded a good approximation of the function. As we would expect from Eq. (7) of the main text, functions with sharper features, i.e. a larger maximal derivative, require a smaller decay rate κ to be approximated well.

II. EXPRESSIVITY

A. Number of independent parameters

Here we expand on the discussion on the number of independent training parameters we obtain with our approach (see Methods). We emphasize that all of these trainable parameters are entirely physical in contrast to hybrid approaches, including the recent works [5, 6], that rely on training digital networks acting on the output.

In general, a scattering matrix element is of the form $S_{j,\ell}(0) = 1 + i\kappa_j(-1)^{j+\ell} \frac{\det H^{(j,\ell)}}{\det H}$. For a neuromorphic system with multi-dimensional input and output, both numerator and denominator automatically contain terms such as x_1x_2 , x_2x_3 and $x_1x_2x_3$ even at $R=1$ with some coefficients defined through the other system parameters. In fact, with D the input dimension

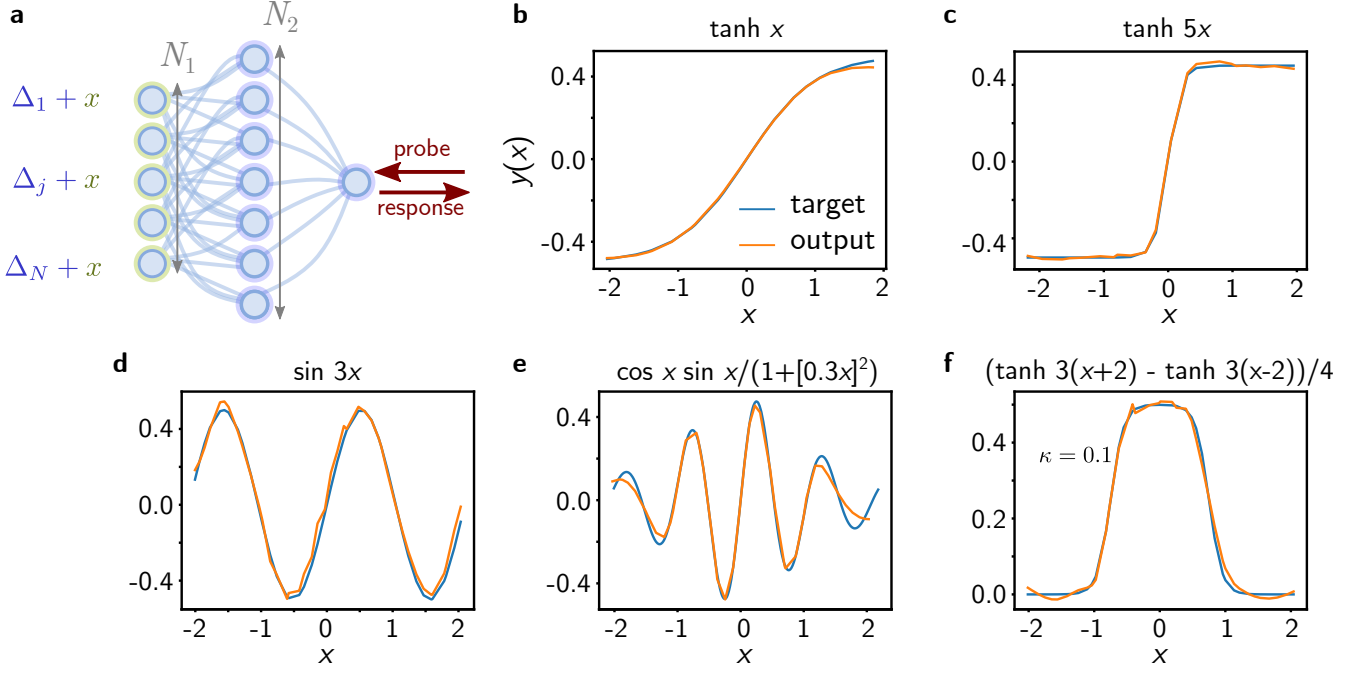


FIG. S2. **One-dimensional function fitting.** Examples for one-dimensional functions fitted with the network shown in **a**. The input enters as offset to the detuning of the first layer and enters $R = N_1$ number of times. A second layer of variable size is connected to a single site. The imaginary part of the scattering response at that site serves as output. The parameters used in the fits were **b** $N_1 = 20, N_2 = 20, \kappa = 1$; **c** $N_1 = 60, N_2 = 30, \kappa = 0.2$; **d-f** $N_1 = 60, N_2 = 30, \kappa = 0.1$. Note that functions with sharper features require larger R and larger N_2 . Typically, κ has to be small to capture sharp features, although, this can to some extent be compensated by rescaling the range of input values by $1/\kappa$.

and R the number of input replications, the number of these terms grows as

$$\sum_{j=1}^{DR} \binom{DR}{j} = 2^{DR} - 1. \quad (\text{S7})$$

However, while, in general, the number of terms grows exponentially with the input dimension and input replication, not all of these coefficients are independent parameters that can be controlled by adjusting the system parameters. The number of independent parameters is typically much smaller.

To find the actual number of independent parameters we can maximally achieve for a neuromorphic system with input dimension D , input replication R and output dimension N_{out} , including systems with the three-layer architecture of Fig. 2 **a**, we consider the map to the minimal reduced system which still contains all input sites and probe sites, see Fig. S1 **d**. The number of couplings entering in this system is equal to the maximal number of independent parameters we can achieve for this given input dimension D and input replication R . We find that the minimal system consists only of input and output modes with all-to-all couplings between them, see Fig. S1 **d**. The total number of couplings in this system is

$$N_{\text{max}} = \frac{(DR + N_{\text{out}})(DR + N_{\text{out}} + 1)}{2}. \quad (\text{S8})$$

In the three-layer architecture, $N_1 = DR$ and $N_3 = N_{\text{out}}$. In addition to that, we have $DR + N_{\text{out}}$ local terms (decay rates, detunings).

Next, we consider the three-layer architecture, Fig. 2 **a**, and we ask how large the second layer N_2 has to be to have enough independent parameters to control all of these effective couplings independently. (Note that the local parameters are automatically independent parameters so we only need to consider the couplings in this derivation.) The total number of couplings between layers of sizes N_1, N_2 and N_3 is given by

$$N_{\text{control}} = N_1 N_2 + N_2 N_3. \quad (\text{S9})$$

This is the number of couplings, we have access to. When $N_{\text{control}} < N_{\text{max}}$ we do not utilise the full potential of the architecture, whereas when $N_{\text{control}} > N_{\text{max}}$, some of the couplings are, at least in theory, redundant and the system has a hardware overhead. Nevertheless, this may present an advantage for the training.

The optimal case is when $N_{\text{control}} = N_{\text{max}}$. Equating expressions (S8) and (S9), we find that to be able to independently control all parameters, N_2 should be larger or equal to

$$N_2 \geq \left\lceil \frac{N_1 + N_3 + 1}{2} \right\rceil. \quad (\text{S10})$$

For the system we consider in the main text with $N_1 = 128$ and $N_3 = 10$, we find $N_2 \geq 70$. This explains why the classification accuracy is limited for smaller N_2 and why we achieve the highest classification accuracy for $N_2 = 80$.

B. Remarks about systems encoding functions with multi-dimensional input and output

In the Methods, we derive the expression for the scattering matrix (15) of a neuromorphic system with a layered architecture, Fig. 2 a. The mathematical form of this equation resembles that of the generalised continued fraction (S2) but in matrix form.

In the literature of condensed matter physics, expressions such as Eq. (15) are known as recursive Green's functions [7] and have been employed to study various condensed matter systems. It is expected that for larger depth L , these networks display localisation when subject to disorder in the couplings and detunings [8, 9]—this may possibly impede the training of such networks (correlations between input and output layer are expected to decrease with L in which case the gradients will also become small). For most applications, $L = 3$ or 4 may already be sufficient. If the application requires deeper networks, introducing gain, e.g., in the form of two-mode or single-mode squeezing, or, alternatively, introducing a few strategic couplings of longer range between layers that are further apart, may resolve this issue of localisation.

C. The role of input replication

An important result described in the main text is that input replication increases the expressivity of the neuromorphic system. Above, we showed this explicitly for one-dimensional functions. This statement is true more generally and indeed also holds for other conceivable, neuromorphic systems based on linear physics, such as the recent works [5, 6]. In general, the number of times R the input enters via system parameters can be thought of as the order of approximation of the non-linear output function. Specifically, a linear physical system always establishes a linear map $S(\omega, \mathbf{x}, \theta)$ between probe and response (in our case, this is realised via the scattering matrix but in other setups S can more generally stand for any linear map facilitated by the linear system, e.g. a Green's function). If the same data input x_j is encoded in multiple parameters, the probe signal can ideally interact with the input data via system parameters at least R times, and therefore S can now not only depend on x_j but also on higher orders $x_j^2, \dots, x_j^R$ etc.

Our proposed setup has an additional advantage: waves propagate back and forth through the system also visiting multiple times the resonators whose detunings encode the data input. The interference between all those paths gives rise to the scattering matrix. Hence, in our approach, even without physically repeating the data, we can already profit from the fact that the signal interacts with the data many times such that the scattering matrix contains higher order terms of the input vector. We see this explicitly from the scattering matrix $S(\omega) = \mathbb{1} - i(\omega\mathbb{1} - H)^{-1}$. Concretely, we split the dynamical matrix H into its diagonal part $\Delta - i\frac{\kappa}{2}$ of detunings and decay rates, respectively, and off-diagonal part J , i.e. the couplings between all modes (not to be confused with the coupling matrices in the layered architecture discussed in the main text), $H = \Delta - i\frac{\kappa}{2} + J$. The output signal in our case is then related to the probe signal via the scattering matrix

$$\begin{aligned} \mathbf{a}_{\text{res}} &= (\mathbb{1} - i\sqrt{\kappa}(\omega\mathbb{1} - H)^{-1}\sqrt{\kappa}) \mathbf{a}_{\text{probe}} \\ &= \left(\mathbb{1} - i\sqrt{\kappa} \left(\left[\omega\mathbb{1} - \Delta + i\frac{\kappa}{2} \right] - J \right)^{-1} \sqrt{\kappa} \right) \mathbf{a}_{\text{probe}} \\ &= \left(\mathbb{1} - i\sqrt{\kappa} \sum_{n=0}^{\infty} [\chi(\omega)J]^n \chi(\omega)\sqrt{\kappa} \right) \mathbf{a}_{\text{probe}} \end{aligned} \quad (\text{S11})$$

with $\chi(\omega) \equiv (\omega\mathbb{1} - \Delta + i\frac{\kappa}{2})^{-1}$. In the last step, we used the Woodbury matrix identity to expand the matrix inverse into the sum. Eq. (S11) reveals the scattering that occurs within the system: waves can scatter off the different neuron modes at which the susceptibility $\chi(\omega)$ modifies the signal and the matrix J connects these different scattering events.

In contrast, imagine that we construct a different neuromorphic system that only makes use of sequential linear scattering in which the input data is encoded in the scattering potential, e.g. by scattering off a spatial light modulator or digital micromirror device, alongside trainable parameters, such as in Ref. [5]. While this also allows for input replication, it is straightforward to see that the meaning of input replication is different from input replication in our setting. To be concrete, the vector of output

signals $\mathbf{a}_{\text{res}}$ as a function of the probe field $\mathbf{a}_{\text{probe}}$ after passing through such a sequential scattering setup can be expressed as

$$\mathbf{a}_{\text{res}} = G_N T_N G_{N-1} T_{N-1} \cdots G_1 T_1 G_0 \mathbf{a}_{\text{probe}} \quad (\text{S12})$$

in which G_j denotes the Green's function which describes the propagation between scattering step j and $j + 1$ (with G_N describing the propagation to the output) and T_j describes the scattering off the scattering potential at the j th setup. Here, the input data and training parameters can be encoded in T_j . If the input data enters only once, e.g. in T_1 , Eq. (S12) only yields a fully linear map of the input data, whereas in Eq. (S11) the map would already be non-linear. This is even more striking when we direct our attention to encoding correlations between different components of the input vector. As we write in the main text, our approach encodes correlations such as $x_j x_\ell$, $x_j x_\ell x_m$ even at $R = 1$, which is also apparent from Eq. (S11). For instance, $R = 1$ is sufficient to obtain a good accuracy on the Fashion-MNIST dataset (Methods). Sequential scattering, however, does not automatically encode correlations if the input enters only once and *always* requires input replication to increase the expressivity beyond that of a linear map.

III. DERIVATION OF THE GRADIENT FORMULAS

Here, we derive the derivative of the scattering matrix w.r.t. a parameter θ_j , Eq. (6),

$$\frac{\partial S(\omega, \mathbf{x}, \theta)}{\partial \theta_j} = -i\sqrt{\kappa} \frac{\partial}{\partial \theta_j} [\omega \mathbb{1} - H(\mathbf{x}, \theta)]^{-1} \sqrt{\kappa}. \quad (\text{S13})$$

The derivative of the matrix inverse M^{-1} w.r.t. a parameter follows from

$$\begin{aligned} 0 &= \frac{\partial}{\partial \theta_j} \mathbb{1} = \frac{\partial}{\partial \theta_j} (M^{-1}(\theta) M(\theta)) \\ &= \left(\frac{\partial}{\partial \theta_j} M^{-1}(\theta) \right) M(\theta) + M^{-1}(\theta) \left(\frac{\partial}{\partial \theta_j} M(\theta) \right). \end{aligned} \quad (\text{S14})$$

Solving for $\frac{\partial}{\partial \theta_j} M^{-1}(\theta)$, we obtain

$$\frac{\partial}{\partial \theta_j} M^{-1}(\theta) = -M^{-1}(\theta) \left(\frac{\partial}{\partial \theta_j} M(\theta) \right) M^{-1}(\theta). \quad (\text{S15})$$

Using this expression for Eq. (S13), we obtain

$$\frac{\partial S(\omega, \mathbf{x}, \theta)}{\partial \theta_j} = i\sqrt{\kappa} [\omega \mathbb{1} - H(\mathbf{x}, \theta)]^{-1} \frac{\partial H(\mathbf{x}, \theta)}{\partial \theta_j} [\omega \mathbb{1} - H(\mathbf{x}, \theta)]^{-1} \sqrt{\kappa} \quad (\text{S16})$$

which with $G(\omega, \mathbf{x}, \theta) = (\omega \mathbb{1} - H(\mathbf{x}, \theta))^{-1}$ leads to Eq. (6) of the main text, $\frac{\partial S_{r,p}}{\partial \theta_j} = i\kappa_p \left(G \frac{\partial H}{\partial \theta_j} G \right)_{r,p}$.

To compute the gradients for the proposed implementation with coupled resonators, we require the derivatives of the dynamical matrix H w.r.t. the detunings Δ_j and the coupling strengths $J_{j,\ell}$. Recalling that $H_{j,\ell} = J_{j,\ell}$ for $j \neq \ell$ and $H_{j,j} = -i\frac{\kappa_j + \kappa'_j}{2} + \Delta_j$, we find

$$\frac{\partial H_{m,n}}{\partial \Delta_j} = \delta_{m,j} \delta_{n,j}, \quad (\text{S17})$$

$$\frac{\partial H_{m,n}}{\partial J_{j,\ell}} = \delta_{m,j} \delta_{n,\ell} + \delta_{m,\ell} \delta_{n,j} \quad (\text{S18})$$

in which the only assumption was that the couplings $J_{j,\ell}$ are real. Together with Eq. (6) we obtain Eqs. (7) and (8).

Note that the measurement of gradients via the scattering matrix always requires knowledge of the phase of the scattering matrix element, i.e., it is not sufficient to measure amplitudes. Specifically, one might be tempted to consider $y_r = |S_{r,p}|^2$ instead of Eq. (4) as the output of the neuromorphic system. While this simplifies the measurement of the output, the gradients still involve the real and imaginary part of the scattering matrix elements.

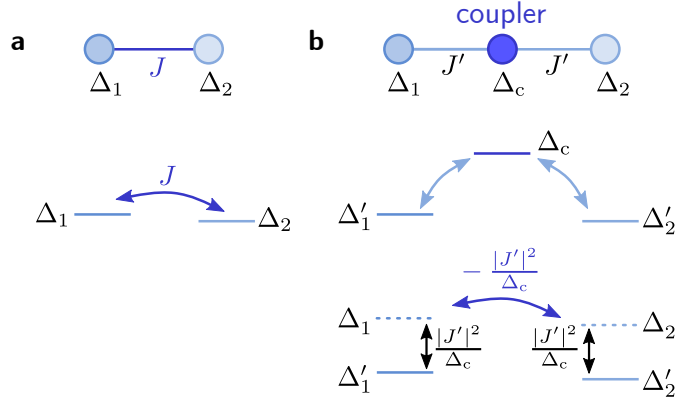


FIG. S3. **Effective coupling mediated by coupler modes.** In one proposal of Fig. 3 **a** we replaced **a** the tunable couplings of strength J by **b** a coupler mode at tunable detuning Δ_c coupled to two neuron modes at fixed coupling strength J' . This gives rise to an effective coupling between the two neuron modes at strength $J = |J'|^2 / (i\frac{\kappa_c}{2} + \omega - \Delta_c)$, Eq. (S21), so the effective coupling strength J can be tuned by tuning Δ_c . At the same time, the detuning of each mode is shifted by the same amount. To map back to exactly the same situation as in **a**, this shift needs to be compensated when tuning Δ'_j , Eq. (S22).

IV. FURTHER DETAILS ABOUT THE IMPLEMENTATION

A. Derivation of the effective coupling

In Fig. 3 we briefly mentioned the option to use coupler modes mediating the coupling between two neuron modes. Here, we derive how a change in the detuning of the coupler mode changes the effective coupling between two neuron modes. The aim is to achieve an effective, predominantly coherent coupling between two neuron modes which can be tuned by adjusting the detuning of the coupler mode, see Fig. S3. This allows to transfer all results obtained for a network with tunable coherent coupling strengths to this system with coupler modes.

For simplicity consider the two neuron modes a_1 and a_2 at detunings Δ_1 and Δ_2 , respectively, coupled via coupler mode c at detuning Δ_c and experiencing decay at κ_c to which both a_1 and a_2 are coupled at strengths J' . The corresponding, Fourier transformed equations of motion are given by

$$\begin{aligned} -i\omega a_1 &= -\left(\frac{\kappa}{2} + i\Delta'_1\right) a_1 - iJ'c \\ -i\omega a_2 &= -\left(\frac{\kappa}{2} + i\Delta'_2\right) a_2 - i(J')^*c \\ -i\omega c &= -\left(\frac{\kappa_c}{2} + i\Delta_c\right) c - i(J')^*a_1 - iJ'a_2. \end{aligned} \quad (\text{S19})$$

Solving for $c(\omega)$, we find

$$c = \frac{J'}{i\frac{\kappa_c}{2} + \omega - \Delta_c} (a_1 + a_2) \quad (\text{S20})$$

which we insert in Eqs. (S19)

$$\begin{aligned} -i\omega a_1 &= -\left(\frac{\kappa}{2} + i\Delta'_1\right) a_1 - i\frac{|J'|^2}{i\frac{\kappa_c}{2} + \omega - \Delta_c} (a_1 + a_2) \\ -i\omega a_2 &= -\left(\frac{\kappa}{2} + i\Delta'_2\right) a_2 - i\frac{|J'|^2}{i\frac{\kappa_c}{2} + \omega - \Delta_c} (a_1 + a_2). \end{aligned} \quad (\text{S21})$$

There are two key points to notice: (i) the coupler mode induces effective coupling between the two modes in which the coupling strength is complex, i.e., it results in both coherent and dissipative coupling. To come as close as possible to an effectively coherent coupling, we require $\kappa_c \ll \Delta_c$, in which case the coherent coupling strength is approximately $-|J'|^2/(\Delta_c - \omega)$. (ii) The coupler modes lead to an on-site shift in detuning and contributes towards the dissipation. To compensate for this detuning shift, we offset Δ'_j by the same amount, i.e.,

$$\Delta'_j = \Delta_j + |J'|^2/\Delta_c. \quad (\text{S22})$$

This gives rise to

$$\begin{aligned} -i\omega a_1 &= -\left(\frac{\kappa}{2} + i\Delta_1\right) a_1 + i\frac{|J'|^2}{\Delta_c} a_2 \\ -i\omega a_2 &= -\left(\frac{\kappa}{2} + i\Delta_2\right) a_2 + i\frac{|J'|^2}{\Delta_c} a_1 \end{aligned} \quad (\text{S23})$$

which are the equations describing the scenario depicted in Fig. S3 a with $J = -|J'|^2/\Delta_c$. Note that if the neuron modes are coupled to multiple coupler modes, the detuning shifts from all coupler modes have to be compensated, so, more generally,

$$\Delta'_j = \Delta_j + \sum_{\ell} |J'|^2/\Delta_{\ell,c} \quad (\text{S24})$$

in which ℓ sums over all coupler modes that mode j is connected to.

In practise, a neuron mode can be connected to a large number of coupler modes, so to avoid large detuning shifts (which may then even be outside the tuning range), one can alternate positive and negative detuning of the coupler modes $\Delta_{\ell,c}$. In that case, most of the terms in Eq. (S24) balance out and the net shift is small.

B. Creating the initial configuration in the proposed implementation

To start training the neuromorphic system, it is first necessary to calibrate the setup, so any fabrication disorder should not be larger than the tunable range, see Fig. 3 d. Once the setup is calibrated, one can fine tune the detunings to a suitable initial configuration. We found in our simulations that starting from configurations in which the neuron modes are effectively approximately resonant (up to some small disorder), i.e., taking into account the detuning shifts (S24) due to the coupler modes, leads to a faster and reliable convergence of the training. The detunings of the coupler modes should be large enough to enable sufficiently strong coupling between neuron modes. In particular, $\Delta_c/\kappa_c \gg 1$ ensures that the coherent coupling dominates over the dissipative coupling induced by the coupler mode, Eq. (S19). In addition, some small disorder in the effective coupling strengths helps to avoid getting stuck early in the training.

In numerical experiments, it has proven helpful to generate a random initial configuration in the ersatz description, i.e., without the coupler modes and to translate this back to a description in terms of coupler modes, i.e., solving for Δ_c . Subsequently, one chooses alternating sign for the coupler mode detunings and calculates the resulting on-site shift (S24) which is then compensated for by changing the neuron mode detuning which effectively makes the neuron modes resonant again. This serves as initial configuration for the training. Subsequent training steps can then be performed *in situ* based purely on the measured scattering matrix.

C. Simulation of the implementation with coupler modes

While it may be preferable to use directly tuneable couplings between neuron modes, one can alternatively, as discussed above, use tuneable coupler modes which provide a second-order coupling. To investigate this scenario, we have simulated an implementation of the network with $N_2 = 80$ from the previous section in the physical architecture discussed here using the ersatz description illustrated in Fig. 3 a. This ersatz description allows us to translate the values of the couplings, Fig. 3, to the corresponding values for the detunings of coupler and neuron modes. In this way, starting from an optimised solution, we achieve a test accuracy of 94.0% which could be further improved by increasing the ratio Δ_c/κ_c or by further training. Alternative to this approach, one can train the system starting from random detunings.

Fig. S4 shows the resulting distribution of the detunings. They spread over a range of approximately $|\Delta|/\kappa \sim 300$, $|\Delta_c|/\kappa_c \sim 800$. While current experiments achieve tuning ranges of $|\Delta|/\kappa \sim |\Delta_c|/\kappa_c \sim 50$ [10], recent proposals suggest that with further optimization the Q-factor could be improved to achieve ratios as high as $|\Delta|/\kappa \sim |\Delta_c|/\kappa_c \sim 2000$ [11].

The large ratio $|\Delta_c|/\kappa_c$ is necessary to overcome the dissipation induced by each of the coupler modes. While the effective coherent coupling induced by a coupler mode is given by $-\frac{J^2}{\kappa_c} \frac{\Delta_c/\kappa_c}{1+\Delta_c^2/\kappa_c^2}$, the contribution of a coupler mode to the dissipation rate of each neuron mode it is coupled to is given by $\frac{J^2}{\kappa_c} \frac{1}{1+\Delta_c^2/\kappa_c^2}$. Assuming each neuron mode is coupled to N_c coupler modes, the ratio between coherent coupling and added local dissipation scales as $\frac{\Delta_c/\kappa_c}{N_c}$. For a high expressivity, it is important that the coherent coupling exceeds the local dissipation therefore requiring large Δ_c/κ_c . Hence, this approach requires either (i) each neuron mode to only be coupled to few coupler modes, e.g., as in a convolution network or other restricted architectures, (ii) tuning ranges of multiple hundreds of linewidths, e.g., through high Q-factors as in the proposal [11], or (iii) means to compensate for the added dissipation, e.g., by introducing a gain mechanism.

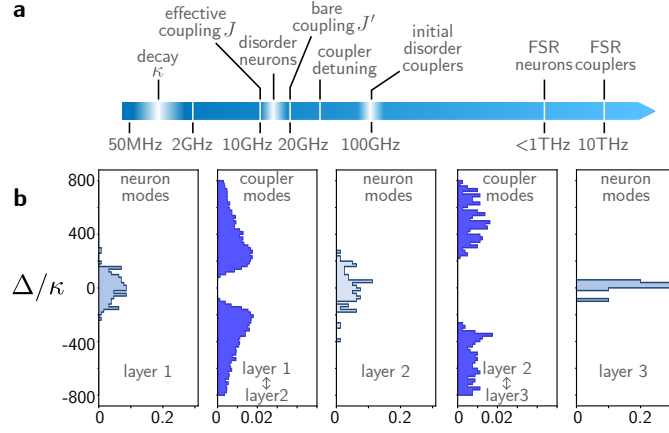


FIG. S4. **Frequencies in the implementation with coupler modes.** **a** Relevant frequencies for the implementation with coupler modes. For a high expressivity, we require a large ratio Δ_c/κ_c . **b** Frequency distribution of a trained neural network with $N_2 = 80$.

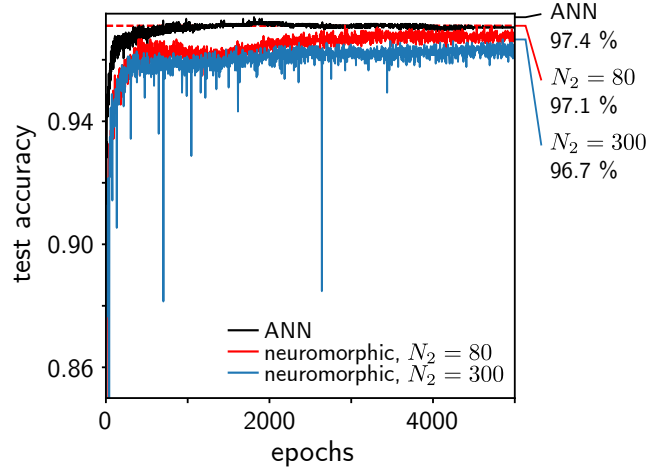


FIG. S5. **Test accuracy for a system with hidden layer size $N_2 = 300$.** The neuromorphic system with $N_2 = 300$ trains more slowly than the system with $N_2 = 80$ indicating that over-parameterisation may impede training. The maximal accuracy for the neuromorphic system continues to increase beyond the 5,000 epochs shown here to a peak accuracy of 96.9% after 9690 epochs.

V. FURTHER DETAILS ON NUMERICAL RESULTS

A. Digit classification with a larger hidden layer

In Fig. 2 b of the main text we show test accuracies for systems with hidden layer sizes up to $N_2 = 80$. As we derived before, the maximal number of training parameters is already achieved at $N_2 = 70$, so, in principle, adding more neuron modes to the hidden layer while keeping the input replication R fixed only over-parameterises the system.

In Fig. S5 we compare the test accuracy of a system with a hidden layer of $N_2 = 300$ with the performance of the system with $N_2 = 80$ which attained the best accuracy and fastest convergence in the main text. We see that although the larger system possesses a sufficient number of parameters, the system does not reach the same test accuracy indicating that excessive over-parameterisation may in fact impede, or slow down, training. The artificial neural network (ANN) with the comparable number of parameters (achieved at a hidden layer size of 560) instead performs slightly better than the ANN with the same number of parameters as the neuromorphic system of $N_2 = 80$, see Fig. 2 b.

B. Classification results for Fashion-MNIST

In Fig. 2 of Methods, we display the test accuracy on Fashion-MNIST during the training. We list the best attained test accuracies in Tab. I. Furthermore, we compared our results to fully linear neural networks (with a softmax function applied at

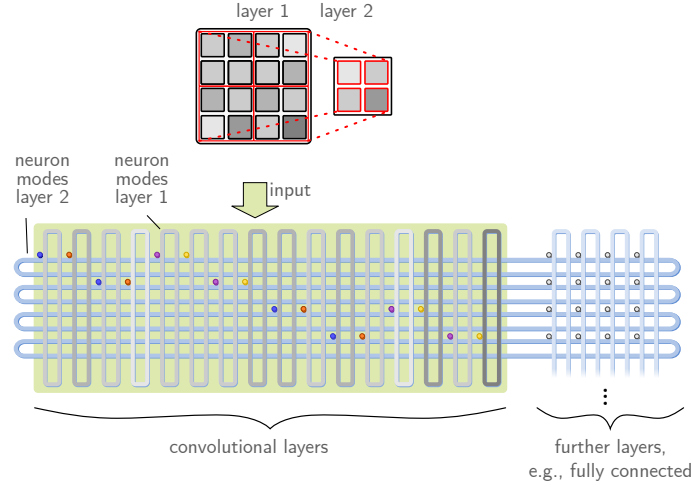


FIG. S6. **Convolutional layer in a possible implementation with racetrack resonators.** Convolutional layers apply a kernel to a two-dimensional image (top). In the implementation, the two-dimensional array of pixels is flattened into a one-dimensional vector. Since the Kernel (here, for simplicity, a 2x2 Kernel is sketched) only acts on a small number of pixels, the coupling to the next layer is very sparse. This is reflected in the reduced number of required tunable couplings (coloured dots). Different colours correspond to different values of the couplings. For simplicity, in this illustration, layer 1 and 2 each only have one channel.

the output), for the various architectures already listed in the table (in the following, the two values denote test accuracies for learning rates $10^{-3}/10^{-4}$): fully connected large: 86.5 % / 86.5 %, small: 85.6 % / 85.5 %; convolutional large: 86.0 % / 86.7 %, small: 85.6 % / 85.9 %.

TABLE I. **Comparison of test accuracy, architecture and number of parameters for different neuromorphic and digital neural networks.** We trained various fully-connected layer and convolutional setups with both our neuromorphic system on either large 28x28 or small 14x14 images as well as artificial neural networks. In all cases we train a categorical cross entropy cost function.

	lr= 10^{-3}	lr= 10^{-4}	architecture	# neuron modes / neurons	# parameters
fully connected:					
neuromorphic, large	0.898	0.895	784 - 784 - 10	1,578	624,074
neuromorphic, small	0.898	0.898	196 - 196 - 10	402	40,778
ANN, large	0.907	0.911	784 - 784 - 10	1,578	623,290
ANN, small	0.904	0.904	196 - 196 - 10	402	40,582
convolutional:					
neuromorphic, large	0.882	0.882	(28x28,1) - (14x14,10) - (7x7,16) - 100 - 10	3,638	85,758
neuromorphic, small	0.878	0.879	(14x14,1) - (7x7,10) - (3x3,16) - 100 - 10	940	19,060
ANN, large	0.905	0.912	(28x28,1) - (14x14,10) - (7x7,16) - 100 - 10	3,638	82,256
ANN, small	0.897	0.906	(14x14,1) - (7x7,10) - (3x3,16) - 100 - 10	940	18,256

VI. CONVOLUTIONAL SETUPS

For conventional ANNs, it is known that for important tasks like image recognition one can gain efficiency by exploiting translational symmetry, using a convolutional neural network. This can drastically reduce the number of trainable parameters, while retaining the same performance or even sometimes resulting in better performance (since the sample efficiency of training increases). We can implement the same idea in our neuromorphic platform. This means that the number of required physical couplings between neuron modes of subsequent layers is significantly reduced, easing the fabrication challenge, Fig. S6. As an example, take a 1D convolution of the form $y'_j = \sum_j K(j' - j)x_j$, where K is the kernel and only has a finite range δj . A conceptually analogous neuromorphic setup in our case would introduce couplings only between neuron modes j' and j in subsequent layers that are separated by no more than δj . Thus, if N is the number of modes in each of these layers, the amount of physical couplers would scale no longer quadratically (N^2) but rather linearly ($N\delta j$). Furthermore, during training updates we would impose the constraint that all the couplings with the same distance $j' - j$ are assigned the same value. Effectively, one would shift the previously existing joint value by the average over all the gradients obtained at any of the couplers assigned to

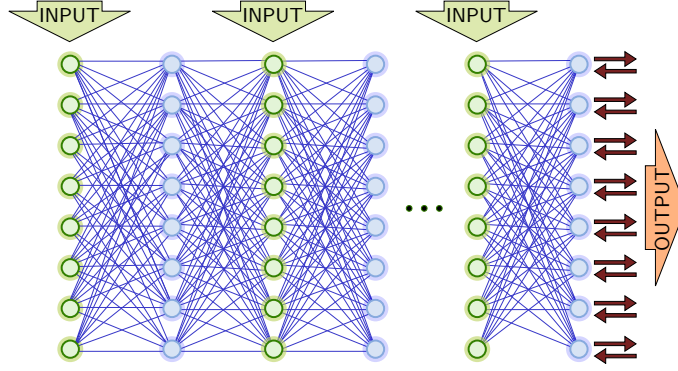


FIG. S7. **Deep neuromorphic system.** The input is repeated across different layers to achieve more complicated non-linear maps and a higher expressivity.

this value, ensuring that all coupling values of this type remain the same. Furthermore, in convolutional ANNs, there exists the concept of channels, i.e., the convolution is of the type $y_{j',c'} = \sum_{j,c} K(j' - j, c', c) x_{j,c}$ with c, c' the indices of the channels. Again, this can be translated to the present neuromorphic platform, provided that we adopt some suitable mapping to the physical neuron mode index n , e.g. $n(j, c) = j + cM$. Here M is the total number of “pixels” in the input, such that the total number of modes in the layer would be $N = Mn_{\text{channels}}$, where the number of channels must be chosen. In general, layers can have different numbers of channels. The biases (detunings) can also be chosen to have the same values for all neuron modes of the same channel in a given layer, though in the numerical experiments reported above we allowed them to vary (this only increases the total number of parameters by a small amount). The extension to high-dimensional convolutions (e.g. 2D, needed for images) is straightforward and has been implemented in our numerical experiments.

VII. EXTENSION TO DEEP NETWORKS

As we discussed in the main text, Methods and in Sec. II of the SI, the system’s linearity imposes an upper bound on the number of independent training parameters for a given input dimension, input replication and output dimension. Increasing the number of trainable parameters (which could be distributed over multiple layers) does not introduce independent parameters although it may help with the training. Therefore, simply increasing the number of layers of the neuromorphic system may not necessarily enhance the network’s expressivity.

In contrast, increasing the input replication R and increasing the number of trainable parameters accordingly (e.g. by introducing more neurons in the hidden layer or more layers), makes the system more expressive. While, in practice, $R = 1$ or $R = 2$ may already be sufficient for applications such as digit recognition or garment recognition (main text and Method), more complex tasks may require other network architectures and in particular deeper networks. In Fig. S7 we sketch the concept for a deep network based on our scheme. The input enters not only in the first layer but distributed over or repeated in subsequent layers. This is expected to further enhance the system’s expressivity. According to Eqs. (13) and (15), this architecture translates to the subsequent application of different non-linear maps similar to a deep neural network. This idea can be further combined with convolutional layers, Sec. VI.

VIII. FLOQUET APPROACH WITH ELECTRO-OPTICAL MODULATORS AND FREQUENCY COMBS

A promising avenue for reducing the number of hardware components is to utilise multiple modes of the same cavities and to couple them through electro-optical modulation. Moving away from the layered architecture described in the main text, here, we discuss a system of $N + 1$ coupled racetrack resonators and N modulators which gives rise to an effective system of N modes with all-to-all couplings. The scheme gives us control over N^2 parameters determining the $N \times N$ scattering matrix which we can use to implement our neuromorphic system.

Taking inspiration from works that generate linear maps via frequency modulation [12–17], we consider the array of racetrack resonators sketched in Fig. S8 a. Each of the L vertical racetrack resonators labelled by index j hosts frequencies $\omega_j^{(\ell)} = \omega_j^{(0)} + \ell\Omega_j^{\text{FSR}}$ with Ω_j^{FSR} the free-spectral range (FSR) of the j th resonator. We assume from now on that the resonator tolerances are small enough so that approximately $\omega_j^{(0)} \approx \omega^{(0)}$ and $\Omega_j^{\text{FSR}} \approx \Omega^{\text{FSR}}$ for all j . The field operator of the ℓ th mode of racetrack

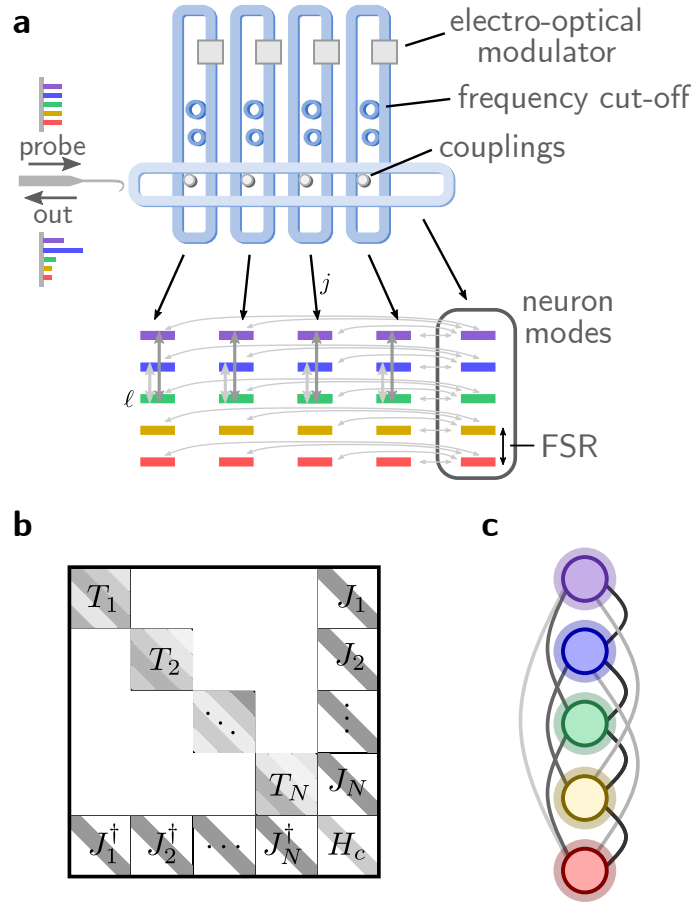


FIG. S8. **Neuromorphic system realised with electro-optical modulation and frequency combs.** **a** Electro-optic modulation achieves coupling between the eigenmodes of racetrack resonators arranged in an array. Each modulated resonator is additionally coupled to two smaller resonators that perturb some of the eigenmodes and thus achieve a frequency cut-off. The modulated resonators are furthermore coupled to an unmodulated racetrack resonator (horizontal) hosting the neuron modes. The modulated resonators effectively realise all-to-all coupling between the neuron modes, see **c**. The scattering response should now be evaluated at the frequencies of the neuron modes which can be achieved e.g. with a frequency comb. **b** Illustration of the dynamical matrix of the full system, see Eq. (S26). The translational invariance brought about by the time-modulation results in the block Toeplitz structure with Toeplitz matrices T_j . Here, $H_c = -i \frac{\kappa_c + \kappa_c'}{2}$.

resonator j is denoted by $b_j^{(\ell)}$. Electro-optical modulation induces coupling between the modes of the form [14]

$$\dot{b}_j^{(\ell)} = -\frac{\kappa_j^{(\ell)}}{2} - i \sum_{j=1}^M (g_j^{(\ell)} b_{j-\ell} + g_j^{(-\ell)} b_{j+\ell}) \quad (\text{S25})$$

in which $g_j^{(\ell)}$ is the coupling induced by the electro-optical modulation of the dielectric constant [14] and M denotes the number of modulation tones (corresponding to the coupling range). Following the idea presented in Ref. [14], a cut-off in this, in theory, infinite frequency ladder can be achieved by coupling to two smaller resonators (one for each cut-off) which perturbs the frequency above and below the desired range shifting them out of resonance with the modulation tones. As a result, we obtain a finite frequency ladder in which N modes are coupled via the electro-optical modulation.

In addition, we couple all racetrack resonators to the same (unmodulated) racetrack resonator hosting modes $c^{(\ell)}$ (the neuron modes) at frequencies $\omega_c^{(\ell)} = \omega_0 + \ell \Omega_{\text{FSR}}$. The coupling between mode ℓ of resonator j and mode $c^{(\ell)}$ is denoted by $J_j^{(\ell)}$. Note that these couplings do not have to be tunable for our scheme and can be assumed as fixed. As we will now show, this additional coupling effectively induces all-to-all coupling between modes $c^{(\ell)}$ of the unmodulated racetrack resonator, Fig. S8 c.

The system is probed by injecting signals $c_{j,\text{probe}}$ at the unmodulated racetrack resonator, see Fig. S8 a, inducing extrinsic losses $\kappa_c^{(\ell)}$ on the ℓ th mode in addition to the intrinsic losses $\kappa_c^{(\ell) \prime}$. For the probe injection, frequency combs can be used which promise high processing speeds and are starting to become integrable in photonic chips [18]. This scheme has the advantage that the entire scattering matrix can be measured with a *single* measurement.

With $\mathbf{b}_j$ denoting the vector of modes of the j th modulated resonator and $\mathbf{c}$ the vector of modes of the unmodulated racetrack resonator, the equations of motion can be cast into the form

$$\begin{aligned} -i\omega \mathbf{b}_j &= -iT_j \mathbf{b}_j - iJ_j \mathbf{c}, \\ -i\omega \mathbf{c} &= -\frac{\kappa_c + \kappa'_c}{2} \mathbf{c} - i \sum_j J_j^\dagger \mathbf{b}_j - \sqrt{\kappa} \mathbf{c}_{\text{probe}}. \end{aligned} \quad (\text{S26})$$

Here, the decay rates $\kappa_c^{(j)}$ and $\kappa_c^{(j) \prime}$ are collected on the diagonals of matrices κ_c and κ'_c , respectively. $J_j \equiv \text{diag}(J_j^{(1)}, \dots, J_j^{(N)})$ describes the coupling to the unmodulated racetrack resonator, and the matrix T_j denotes the Toeplitz matrix

$$(T_j)_{m,n} = \begin{cases} g_j^{(n-m)} & : m \neq n \\ -i\frac{\kappa_j^{(m)}}{2} & : m = n \end{cases} \quad (\text{S27})$$

that accounts for the translational invariance of Eq. (S25) w.r.t. $j - \ell$. $\kappa_j^{(m)}$ is the dissipation rate of the m th mode in the j th resonator. Together, the dynamical equations (S26) define the dynamical matrix H illustrated in Fig. S8 b governing the dynamics of $(\mathbf{b}_1^T, \dots, \mathbf{b}_L^T, \mathbf{c}^T)^T$.

Solving for Eqs. (S26) for $\mathbf{b}_j$ and inserting the expression into the equation for $\mathbf{c}$, we obtain

$$-i\omega \mathbf{c} = -\frac{\kappa_c + \kappa'_c}{2} \mathbf{c} - i \sum_j J_j^\dagger (\omega \mathbb{1} - T_j)^{-1} J_j \mathbf{c} - \sqrt{\kappa} \mathbf{c}_{\text{probe}}. \quad (\text{S28})$$

This calculation reveals that the modulated racetrack resonators effectively engineer coupling of range M between all N modes $c^{(\ell)}$ of the unmodulated resonator, Fig. S8 c. With $M = N$ modulation tones, this amounts to all-to-all coupling. Together with the input-output relation [19, 20] $\mathbf{c}_{\text{res}} = \mathbf{c}_{\text{probe}} + \sqrt{\kappa_c} \mathbf{c}$, we calculate the scattering matrix relating $c_{j,\text{probe}}$ to $c_{j,\text{res}}$

$$S(\omega) = \mathbb{1} - \sqrt{\kappa} [\omega \mathbb{1} + i\frac{\kappa + \kappa'}{2} - \sum_j J_j^\dagger (\omega \mathbb{1} - T_j)^{-1} J_j]^{-1} \sqrt{\kappa}. \quad (\text{S29})$$

The resulting scattering matrix (S29) has dimension $N \times N$. Through the modulation, we obtain control over M parameters in L modulated resonators—in total $M \cdot L$ parameters. Therefore, by setting $M = N$ and $L = N$, we obtain N^2 parameters (the couplings $g_j^{(\ell)}$) defining the $N \times N$ scattering matrix. It was demonstrated numerically in Ref. [17] that a related approach (from a mathematical point of view) allows to generate arbitrary unitary matrices. In our neuromorphic scheme, some of these N^2 parameters can be used to encode the data input, others can serve as trainable parameters.

This approach has the advantage that it is hardware efficient as it requires only $N+1$ racetrack resonators and N electro-optical modulators rather than N^2 electro-optically tunable couplings. Gradients are also easily accessible by attaching waveguides to the modulated racetrack resonators and measuring the frequency-dependent scattering matrix.

IX. SCALABILITY

Neuromorphic architectures will become useful in practical applications only if they scale well towards problem sizes of interest. This requires us to discuss aspects of resource scaling with respect to space, time, and energy consumption. The main point of our particular neuromorphic approach discussed in the main text is to extend linear optical neuromorphic devices towards non-linear information processing via parametric input encoding. For that reason, most aspects of scalability are not peculiar to our approach but actually refer to the wide range of existing linear optical platforms already under consideration in the community. Below we summarise various aspects of scalability with an emphasis on integrated photonics devices, since they seem to offer the greatest flexibility and were therefore chosen as the basis for our discussion in the main text. Beyond that, we highlight the few aspects that arise specifically because of our approach, related to the modulation of parameters for input encoding and training.

A. Space

It is well-known that space requirements impose the strongest limitations on optical neural networks and that they are not directly competitive with electronic devices when this aspect is taken in isolation. It is therefore important to estimate how

the proposed architectures would fare in this regard and to establish that they can reach component numbers that already allow for interesting applications in a reasonable footprint. Once this has been achieved, optical networks can make up for the footprint drawback by allowing for rapid inference, reduced energy consumption, opportunities for novel 3D (or multilayer 2D) structures [21], and the potential for frequency multiplexing (see Sec. VIII). When neurons are implemented as on-chip optical waveguides, both their transverse extent and nearest-neighbor separation are on the order of a wavelength. This means that an array of 1000 neuron waveguides can be arranged in a transverse extent of a few mm. In fact, the major space constraint in an integrated-photonics neuromorphic device is not determined by the layout of the waveguides themselves but rather imposed by the need to have tuneable couplers between these waveguides. Each such coupler may eventually require an area of about $(10\mu\text{m})^2$ when designed carefully [22, 23], e.g., relying on the recent advances of inverse photonic design. This amounts to about 10^6 tuneable connections (learnable weights) in an area of about a square centimeter, enabling already interesting applications. Another space requirement, unique to our approach, pertains to the training phase, where our approach allows to directly read off gradients from physical measurements. To achieve this in a space-efficient way, we have proposed to implement the required measurements of the scattering matrix elements in a way that avoids the need to thread N^2 waveguides into an area containing $N \times N$ coupling elements. Instead, localized measurements are preferred. Grating tap monitors (which redirect some part of the light out of the chip plane, to be recorded on a camera) [24] still need an area of about $50\mu\text{m} \times 20\mu\text{m}$ [22], but in an optimised implementation they could be replaced by integrated photodetectors with much smaller footprint, compatible with the approximately $(10\mu\text{m})^2$ assumed above per coupler. Finally, we need to consider the electrical access needed for electrical tuning of the tuneable couplers. Here, an optimised implementation would not lay out the wires in a 2D geometry, but rather employ multilayer chip structures with electrical through wires for efficient access. Such scalable technologies are currently also being developed and applied for the purposes of controlling quantum computing chips.

B. Timing for inference

Speed is one of the obvious advantages of optical platforms. Much of that advantage is due to the extremely parallel nature of information processing: In a standard digital artificial neural network implementation, every connection contributes one multiply-accumulate operation during inference, and per default all of these operations happen sequentially. This leads to $N_{\text{layer}} \times N^2$ operations when N is the typical number of neurons per layer, for a fully connected network, and $N_{\text{layer}} \times N \times N_{\text{kernel}} \times N_{\text{channels}}^2$ for a convolutional network with a kernel of given size and a typical number of channels N_{channels} . While GPUs and TPUs try to overcome this by offering partially parallel processing, in an optical neuromorphic platform the entirety of these operations automatically happens in parallel for any given layer, such that the total time needed just scales linearly in the layer number N_{layer} , a very significant gain over the usual electronic implementations. One great advantage of repurposing linear optical circuits for non-linear information processing via input encoding is the fact that the whole device does not require any electro-optical conversion for non-linearities, which would otherwise slow down operation significantly. Rather, the device remains passive during inference. Propagation delays (at the speed of light in the waveguide medium) are only about 0.1nsec for 1cm and are therefore not the limiting factor. The main bottlenecks for inference speed in an integrated-photonics device employed in our approach are rather the ringdown rate for any resonators and the coupling strength between those resonators and the rate for input encoding. The bandwidth for all of these is on the order of GHz (see also our separate discussion of currently achievable modulation speeds in the main text). The total estimated propagation time through the device is $N_{\text{layer}} \times 1/J$, with J the effective coupling between neurons. Since multiple scattering of the propagating waves is desired in our approach, this number is still multiplied by a small fixed factor for one complete inference evaluation. Overall, that is on the order of about a nanosecond per layer for the experimental parameters discussed in the main text, independent of the total number of neurons or connections.

C. Timing for training

During training, besides carrying out inference on the training samples we also need to extract gradients. One considerable advantage of the approach proposed here is the ability to efficiently extract gradients via physical observations. For any of the output neurons, we need to obtain the scattering matrix amplitude between that output port and any tuneable optical element inside the whole device. This can be achieved by sending light into this port and extracting all the resulting light amplitudes throughout the device simultaneously (e.g. via grating tap monitors, as described in the main text). Thus, the overall time needed for the process of gradient extraction scales very favourably, as N_{output} , independently of the number of trainable parameters. This has to be compared with the parameter shift method, which can be applied to any neuromorphic device but where the network would be evaluated repeatedly, once for each trainable parameter. In principle, if a reliable model were known, one could train on a digital computer, using the standard backpropagation algorithm. However, due to the significant advantage of the optical neuromorphic platform with regard to inference time, this still will not outcompete the physics-based approach described here. Once the gradient has been observed via optical means, the parameter update has to be applied. This step does not require

any elaborate processing and, in principle, an analog electrical circuit could convert the observed light amplitudes into the parameter update, which would enable fully analog parallel processing. It is also conceivable to employ slightly more advanced methods, such as adaptive gradient optimisers, although this would require converting the signals to digital for storage and processing. The cost for this separate processing step would scale linearly in the number of parameters (modulo parallelisation enabled via use of GPUs etc.), but the processing costs per parameter are relatively small as compared to digital inference or backpropagation.

D. Energy

The energy budget for optical neuromorphic systems of the type considered in our approach scales very differently from that of (opto-)electronic approaches, since most operations (except input encoding and final detection) rely simply on propagation through passive optical components. This means that for these operations there is no energy cost linked by default to the processing. Under these circumstances, the remaining overall energy cost then is determined by two factors: the energy of the light pulses and the energy required for input encoding. The light pulses need to be created and sent through the system and eventually absorbed in the detection ports. To reduce the shot noise level to below 1%, any light pulse sent into one of the probe ports requires about 10^4 photons, which amounts to an energy on the femto-Joule level per such pulse during inference (estimated based on [25]). The number of probe ports is equal to the number of output neurons. Moreover, in our scheme, we assume optical amplitude detection which requires homodyne measurements. The local oscillator reference pulses have to have at least the same energy to retain the same noise level. Any residual absorption is only relevant if it diminishes light levels by such an extent as to affect eventual detection or to generate problematic heating of the device. The typical parameters discussed in the main text have been chosen so that propagation through multiple layers is possible without such appreciable absorption (i.e. intrinsic absorptive decay rates are smaller than coupling rates between different layers by some factor). Even if the energy of all N_{probe} light pulses were absorbed in the device, at GHz processing speeds with $N_{\text{probe}} = 1000$ output neurons that would only amount to mW heat dissipation, which can easily be supported in these devices that do not need to operate at low temperatures anyway. The energy required for input encoding relies on the electrical modulation of optical components, with typical values on the order of several 10s of femto-Joule per bit [25]. Depending on the number of input neurons (each of which requires this modulation), this may be the most relevant energy consumption for inference. Independently, there is also energy consumption during training: the main aspect here is that the measurement of the scattering matrix elements requires the injected light to be measurable at all of the optical elements that correspond to trainable parameters. Correspondingly, to keep the shot noise level down, one needs about a femto-Joule per such trainable parameter.

X. ANALOGUE ELECTRICAL CIRCUITS

The general idea of exploiting the non-linear structure of the Green's function of a linear system transfers to other systems such as analogue electrical circuits. Here, we sketch the idea for a circuit network of resistors, but the same principle holds for more general electric circuits that can, for instance, also contain inductors and capacitors.

Consider an electrical circuit in which resistors $\rho_{j,\ell}$ connect nodes j and ℓ in a resistor network. Kirchhoff's law establishes a relation between the current I_n at each node n and the voltage V_n via the connectivity matrix L or Laplacian of the network. Concretely,

$$-I_n = \sum_m L_{n,m} V_m. \quad (\text{S30})$$

with

$$L_{m,n} = \begin{cases} \frac{1}{\rho_{m,n}} & : m \neq n \\ -\sum_m \frac{1}{\rho_{m,n}} & : m = n \end{cases}. \quad (\text{S31})$$

Our ultimate goal is to calculate the resistance between nodes j and ℓ . To that purpose, we solve the above equation for the voltages, by calculating the circuit's Green's function [26–28]

$$G = (L - \mathbf{v} \cdot \mathbf{v}^T)^{-1} \quad (\text{S32})$$

with $\mathbf{v} = (1, 1, \dots, 1)^T$. This matrix of ones $\mathbf{v} \cdot \mathbf{v}^T$ is the outer product of the eigenvector corresponding to the zero eigenvalue of L which accounts for the fact that adding a constant, uniform potential to every node yields the same circuit with the same relative voltages.

The two-point resistance $R_{j,\ell}$ between nodes j and ℓ is then given through elements of this Green's function

$$R_{j,\ell} = G_{j,j} + G_{\ell,\ell} - (G_{j,\ell} + G_{\ell,j}). \quad (\text{S33})$$

The elements $R_{j,\ell}$ define the *impedance matrix* of the system. Similar as for the scattering matrix, the impedance matrix is a non-linear function of the resistors $\rho_{m,n}$, so we can interpret a set of tunable $\rho_{m,n}$ to be the input $\mathbf{x}$ of our neuromorphic system and some other tunable resistors $\rho_{m',n'}$ to be the learning parameters θ . The output is the measured two-point resistance $R_{j,\ell}$ between a suitable set of nodes. If the ground of the circuit is fixed at a certain node, e.g. node j , the output is proportional to the measured voltage difference to nodes ℓ such that the output can be recorded with a single measurement step. As before, the linear nature of the system enables us to directly measure gradients via the impedance matrix.

XI. COMPARISON TO OTHER MACHINE LEARNING APPROACHES

A. Artificial neural networks

One of several obvious aspects distinguishing our neuromorphic approach from conventional digital artificial neural networks is the non-sequential processing: even when we adopt a layered structure (which is only one particular choice in our scheme), the waves injected at some point do not merely travel sequentially through a succession of layers but instead can experience multiple back-reflections, with all paths interfering to yield the final output. This distinguishes our approach both from artificial (feedforward) neural networks and recurrent networks.

On the other hand, from a mathematical point of view, the scattering matrix that describes the output at the end of the layer architecture in Fig. 2 a of the main text (Eqs. (13) and (15) of the Methods) shows some similarity to the mathematical structure of artificial neural networks: the repeated application of a linear map (facilitated by the hopping) and a non-linear operation (matrix inverse). Due to the linear nature of the system, the attained non-linearities are of a particular form. However, that does not restrict the expressivity which can be enhanced by encoding the same input data in more than one system parameter (Sec. I and Sec. II of the SI). This leads to another difference to artificial neural networks: given input dimension and input replication, there is a bound on the maximal number of independent training parameters (Methods and Sec. II of the SI).

B. Kernel methods

Before the recent large-scale success of neural networks, the most widespread approach in practical machine learning applications was represented by kernel machines, with kernel support vector machines the best-known example [29]. Kernel machines start from the idea of mapping the input into a high-dimensional space, by applying a given set of non-linear functions, called feature maps. Afterwards, a linear classifier may be enough to separate the classes in the high-dimensional space, even though the problem was not linearly separable when phrased in the original low-dimensional input space. Optimizing this final linear step requires evaluation of the overlaps of feature maps, giving rise to the kernel (this helps to avoid actually computing the initial nonlinear feature map when calculating the weights). In the context of neuromorphic systems, reservoir computing relies on the same basic insight [30]. When an input is applied to a non-linear dynamical system in the form of some external driving, the dynamics produces a complex non-linear map of this input to the resulting physical observables. These already non-linearly deformed results, embedded in a higher-dimensional space, can then again be used as input for a linear perceptron, to obtain the final classification or regression result. Both kernel machines and reservoir computing share the characteristic that the original non-linear mapping is fixed, given by the selected feature maps or by the non-linear dynamics. The training parameters only enter into the subsequent linear mapping. I.e., both Kernel methods and reservoir computing rely on the application of a fixed, untrainable non-linear map $N(\mathbf{x})$ on the training data, followed by a trainable linear map $\mathcal{L}_\theta$, such that the output is given by $f_\theta(\mathbf{x}) = \mathcal{L}_\theta(N(\mathbf{x}))$.

In contrast, for our approach, there is no such fixed non-linear mapping from the input to some higher-dimensional space followed by a clearly separate subsequent trainable linear processing step. Rather, the multiple scattering of waves propagating linearly in a physical system creates on its own an output depending non-linearly on the parametrically encoded input, with the training parameters entering this non-linear mapping in an irreducible way, i.e., the scattering matrix $S(\omega, \mathbf{x}, \theta)$ cannot be split into a fixed non-linear map $N(\mathbf{x})$ and a trainable linear map $\mathcal{L}_\theta$.

C. Restricted Boltzmann machines

Data moving back and forth between layers is also a prominent aspect of restricted Boltzmann machines (RBMs) [31], which is why we briefly comment on the differences w.r.t. RBMs as well. The first difference between RBMs and our approach consists

in the goal: RBMs aim to learn to sample from an observed high-dimensional probability distribution, while our scheme aims for the more straightforward tasks of classification and regression. Indeed, the back-and-forth between visible and hidden layer in an RBM is needed to sample from a Markov chain. The aim of modeling a probability distribution also makes training an RBM a bit more involved, leading to contrastive-divergence learning as an approximate way to solve the minimization of a Kullback-Leibler divergence with the desired probability distribution. This learning procedure relies heavily on the restricted architecture of an RBM, while in our supervised training scheme we can evaluate gradients directly from physical observations even for arbitrary architectures (connectivities). RBMs do not naturally have an input, since their aim is just to model a probability distribution, but if one wants to keep some input data, it would simply be fixed in some of the visible neurons (producing a conditional probability distribution) and not be encoded into the parameters of the model. Apart from the differences in the goal, the training, and the input data, there is one physically even more fundamental distinction: We consider coherent interference between waves, while an RBM models an incoherent stochastic process that could be viewed physically like an ensemble of classical particles making random transitions between neurons (with particle creation and annihilation possible processes), according to rates that can be trained.

D. Quantum neural networks

Our approach utilizes input encoding via tuneable parameters as a means to produce non-linearity in an otherwise linear wave system. The general concept of input encoding in parameters can also be found as an ingredient in quantum neural networks (QNNs), so a comparison might be helpful. QNNs are a form of variational quantum circuits, i.e. they consist of a temporal sequence of continuously parametrized quantum gates (unitary transformations) acting on qubits [32]. Most of the gate parameters (often in the form of qubit rotation angles) are trainable, while some are set aside for encoding classical input data. The output is formed by the expectation values of measurements of the qubits in their final state, after the sequence has been run. The hope for QNNs is that the linear dynamics in the exponentially large many-qubit Hilbert space might lead to a quantum advantage over classical information processing, although this vision still has to be demonstrated (even conceptually) for practically relevant applications. The requirements for implementing QNNs are extremely stringent. They rely on the preservation of quantum coherence in a controllable large-scale quantum computing platform, with the overall decoherence rate increasing linearly in the number of qubits. There are strong indications that today's state-of-the-art noisy intermediate scale quantum devices with hundreds of qubits are not sufficient for a quantum advantage in QNNs but rather even much more complex fault-tolerant error-corrected quantum computers will be needed.

In contrast, the classical linear wave scattering devices discussed in our approach can be built today and operate at room temperature. Moreover, reading off the output is straightforward: One pass of a linear wave through the device is enough, while QNNs require many runs of the same gate sequence to suppress the measurement projection shot noise of binary qubit outcomes, which constitutes an important bottleneck for training. Even for the relatively fast superconducting-qubit platform, a QNN would have an inference time for even only a single run of several microseconds, far longer than the nanosecond time-scale we envisage here for light propagation through integrated-photonics devices. While fast light propagation would also be available in photonics-based QNNs, they require feedforward operations and the injection of entangled quantum states at scale, due to the lack of large non-linearities at the few-photon level. Another important conceptual difference between QNNs and our approach is that we rely on multiple scattering throughout a wave device, while QNNs apply a fixed sequence of linear operations sequentially in time. As a consequence, we can obtain complex non-linearities already without any additional input replication. Finally, the availability of physical backpropagation for efficient gradient extraction for all training parameters simultaneously has no counterpart in QNNs.

On a more general level, quantum computers seem to share with our classical approach the idea that linear systems can produce non-linear processing: a quantum algorithm produces a non-linear function of the input even though the quantum dynamics of the wave function are linear. However, this superficial similarity merits closer inspection which reveals important differences. First, we have to distinguish carefully between linearity of physical dynamics and linearity of the mathematical description. Quantum computers are based on physical non-linearities, i.e., typically interactions between qubits, which themselves have to be non-linear physical systems (in unconventional variants of quantum computing the required non-linearities are produced via measurement and feedback). In contrast, the wave scattering systems we consider display linear physical dynamics. Therefore, on the physical level, there is a clear distinction, which is also reflected in the drastically different experimental challenge of building and controlling the non-linear systems for quantum computing vs. the linear platforms we envisage. On the mathematical, formal level the situation is a bit more involved, but eventually also there a clear distinction can be observed: The non-linear quantum system can be described via a linear evolution equation of a wave function, but at the price of working in an exponentially large Hilbert space. A similar situation exists even in classical dynamics: One can choose to describe a non-linear classical system via the linear evolution of its phase space density (in the form of a Fokker Planck equation) or of Koopman operators. Again, this comes at the price of working in an infinite-dimensional space, i.e., mathematically much more complex than the low-dimensional non-linear equations of motion for the physical degrees of freedom. By contrast, a physically linear system (like the one we are considering) allows for a linear mathematical description already in this low-dimensional space of

physical degrees of freedom.

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